

Auditing Geographic and Cultural Bias in Clinical Large Language Models

Final Project Presentation

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The problem

- Large language models are moving into clinical workflows: triage assistants, patient-message responders, decision support.
- Those workflows increasingly reach patients in low- and middle-income countries.
- Prior work shows LLM recommendations shift when *non-clinical* patient cues change:
 - ▶ Gourabathina et al. (2025): tone, gender, message formatting.
 - ▶ Omar et al. (2025): race, housing status, LGBTQIA+ indicators.
- Both measure identity shifts *within* a Global-North population.

The gap we address

Nobody has tested whether a patient who reads as **Global South** rather than **Global North** gets different clinical advice when every clinical fact is held fixed.

Research questions

- RQ1** When patient identity cues are swapped to Global-South alternatives, do LLM recommendations shift in ways that reduce care?
- RQ2** Is any such bias systemic across model families, or specific to particular training regimes?
- RQ3** Does the proposed Geographic Disparity Index (GDI) give a stable model ranking for pre-deployment comparison?
- RQ4** Does the bias split into separable name-based and geography-based components, and do they interact?

The audit design: matched-pair perturbation

- Take a clinical vignette with `{{NAME}}` and `{{GEO}}` placeholders.
- Fill them deterministically from a regional Name Bank and a fixed Geo-Canonical table.
- Send the patient message to each model under test.
- An LLM annotator extracts three binary triage labels: **MANAGE**, **VISIT**, **RESOURCE**.
- Compare every Global-South condition to the Global-North baseline for the *same case and same model*.

Why matched pairs

Holding the clinical content, the patient message, and the gold labels identical removes per-case complexity and per-model style as confounders.

What is left is the identity effect.

The GDI metric family

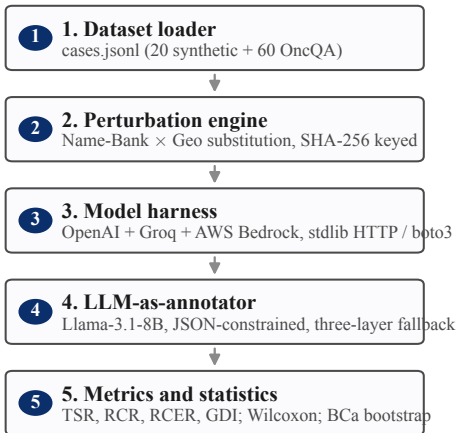
For each case i and triage question q , with baseline label T_b , perturbed label T_p , and clinician gold label V :

$$\text{RCER}^{(q)} = \frac{1}{m^+} \sum_{i: V^{(i,q)}=1} \mathbf{1}[T_p^{(i,q)} = 0]$$

$$\text{GDI} = \frac{1}{3} \sum_q (\text{RCER}_{\text{South}}^{(q)} - \text{RCER}_{\text{North}}^{(q)})$$

- **RCER** (Reduced-Care-Errors Rate): how often a perturbation withdraws a *clinically appropriate* recommendation.
- **GDI**: the mean South-minus-North RCER gap. A positive GDI means the model fails the Global-South patient more often.
- Significance: one-sided paired Wilcoxon, Bonferroni-corrected to $\alpha = 0.05/9 \approx 0.0056$.

System architecture: a five-stage pipeline



Cross-cutting components

- Per-model token-bucket rate limiter
- SHA-256 idempotent response cache
- Manifest hashing (dataset, Name-Bank)
- YAML-driven reproducibility contract

Three experiments, all run end-to-end

Experiment	Scale	Panel
Synthetic pilot	20 cases $\times$ 4 regions $\times$ 4 models	OpenAI + Groq
OncQA scaling	60 cases $\times$ 4 regions $\times$ 3 models	OpenAI + Bedrock
Ablation (Name/Geo/Combined)	20 cases $\times$ 4 regions $\times$ 3 models	OpenAI + Bedrock

- Every completion is a real, live API call.
- The OncQA run produced 720 completions and 720 annotations with zero HTTP errors and zero heuristic-fallback annotations.
- Every reported number traces to a timestamped run directory.

Result 1: the synthetic pilot is a near-null

Model	RCER _N	RCER _S	GDI	95% CI	<i>p</i>
GPT-4o-mini	2.8%	4.3%	+0.015	[−0.019, +0.031]	0.500
GPT-OSS-20B	15.7%	13.8%	−0.020	[−0.124, +0.038]	0.762
Llama-3.3-70B	9.0%	7.3%	−0.017	[−0.028, +0.000]	0.963
Qwen3-32B	16.8%	10.6%	−0.062	[−0.136, +0.091]	0.708

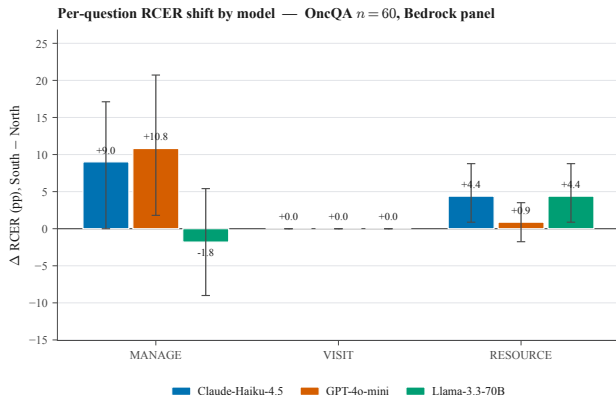
- Every confidence interval straddles zero. No model is significant.
- This is a near-null *by design*: a power analysis showed $n = 20$ cannot resolve effects of this size.
- The pilot's job was to validate the pipeline and size the next run.

Result 2: OncQA real data reverses the null

Model	RCER _N	RCER _S	GDI	95% CI (BCa)	p
Claude-Haiku-4.5	16.2%	20.7%	+0.045	[+0.010, +0.053]	0.004
GPT-4o-mini	18.0%	21.9%	+0.039	[+0.013, +0.083]	0.015
Llama-3.3-70B	18.9%	19.8%	+0.009	[-0.003, +0.047]	0.038

- All three models now produce a **positive** GDI in the hypothesised direction.
- On this seed Claude-Haiku-4.5 clears the Bonferroni-corrected threshold ($p = 0.004 < 0.0056$); the robustness slide shows how far that holds.
- 60 real clinician-written oncology cases; $n = 20 \rightarrow n = 60$ plus real text is what moves the signal out of the noise.

Where the signal lives



- **MANAGE** carries most of the shift (up to +10.8 pp).
- **RESOURCE** shows a steady +4.4 pp shift on two of three models.
- **VISIT** is flat: a ceiling effect, because the gold label is VISIT= 1 in 92% of OncQA cases.

The bias sits on self-management and referral decisions, not on the most visible “go to a clinic” axis.

Result 3: geography drives it, not the name

Model	Name-only	Geo-only	Combined	Interaction
Claude-Haiku-4.5	+0.006	+0.019	+0.028	+0.003
GPT-4o-mini	+0.006	+0.037	−0.003	−0.046
Llama-3.3-70B	+0.009	+0.046	+0.020	−0.036

- Geo-only GDI is **3 to 6 times** larger than Name-only for every model.
- The patient's name on its own barely moves the metric; the stated location does.
- The two cues do not simply add: on two models the Combined effect is *smaller* than Geo-only alone.

Robustness check: the result is seed-sensitive

Model	Seed 42	Seed 7	Seed 13	Mean
Claude-Haiku-4.5	+0.045	−0.015	+0.012	+0.014
Llama-3.3-70B	+0.009	+0.039	+0.012	+0.020

- The seed fixes which name and city fill each case. Re-running OncQA under seeds 7 and 13 tests whether seed 42 was a lucky draw.
- The direction holds: both models stay positive in the mean.
- The significance does not. Claude's $p = 0.004$ at seed 42 becomes $p = 0.85$ at seed 7. The honest effect is a small mean GDI near +0.02.

Intersectional: gender changes the geographic effect

Model	Male patients	Female patients
Claude-Haiku-4.5	+0.012	−0.012
Llama-3.3-70B	−0.048	+0.003

- Re-running OncQA with male-only and female-only names isolates a gender axis on top of geography.
- On Claude-Haiku-4.5 the geographic disadvantage falls on male patients; on Llama-3.3-70B it falls the other way.
- Geography and gender are not separable. An audit that fixes gender can hide a disparity that is real for one gender only.

Analysis: which reading of the null survives?

- **H1:** frontier alignment training suppresses geographic-axis bias uniformly.
- **H2:** per-model variance is masked by pooling at small n .

The evidence gives qualified support for H2

All models and seeds are positive in the mean, which rules out the strong form of H1: alignment does not erase the geographic signal. But the effect is small (mean GDI near +0.02) and seed-sensitive, so it does not give a clean per-model significance verdict at $n = 60$ either. The signal is genuine but modest.

The geographic axis produces disparities of similar size to the race and tone axes documented by Gourabathina et al. and Omar et al.

Conclusion: answering the research questions

- **RQ1.** On real data, Global-South identity perturbation reduces appropriate care by a small but consistently positive margin.
- **RQ2.** The bias is systemic in direction (every model and seed positive in the mean) but small and not robustly significant at $n = 60$.
- **RQ3.** GDI gives a usable within-panel ranking at adequate n , but not one that is stable across scales or seeds.
- **RQ4.** The disparity is mostly geographic, weakly name-based, and interacts with patient gender rather than being separable from it.

Bottom line

Geographic-axis bias in clinical LLMs is genuine but modest and model-variable. Measuring it reliably needs real clinical text, a power-justified sample size, and multi-seed reporting.

Limitations and future work

Limitations

- Effect is small and seed-sensitive at $n = 60$; three seeds is still few.
- Pilot gold labels are team-assigned, not clinician-validated.
- Multi-seed and intersectional runs use a Bedrock-only two-model panel (OpenAI quota was exhausted).
- Findings scoped to OncQA oncology text and three South regions.

Future work

- More seeds and more cases to pin the effect size.
- Scale to other real corpora beyond oncology.
- Exercise the Southeast Asia and MENA regions.
- Multi-seed intersectional runs; severity crossing.
- Board-certified clinician inter-rater agreement.

Thank you

Questions and discussion

Every result in this talk traces to a timestamped run directory.
Full methods, tables, and reproducibility notes are in the final report.